PocketClinic: An On-Device Clinical Assistant Built on a 3B-Parameter Model

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Abstract

PocketClinic is an on-device Medical QA assistant built on a 3B Small Language Model. It is fine-tuned with LoRA-QA and runs fully on a hospital tablet, so patient data leaves neither the device nor the care setting. On MedQA, PocketClinic reaches approximately 80% of CuraLM-70B's accuracy while using a fraction of the memory and latency. The main limitation is safety: it hallucinates drug interactions more often than large models. PocketClinic therefore demonstrates a privacy and deployment advantage, not a replacement for expert review.

Keywords

On-device clinical AI, LoRA-QA, MedQA, Privacy

1. Introduction

Clinical assistants must balance Medical QA quality with privacy, latency, and memory. CuraLM-70B [1] is accurate but requires data-center hardware. LoRA-QA [2] offers a parameter-efficient adaptation route, while structured relational memory [3] suggests a future way to reduce fabricated drug interactions. PocketClinic focuses on offline execution and explicitly leaves external grounding for future work.

2. Method

PocketClinic begins with a 3B Small Language Model and applies LoRA-QA on curated medical question-answer data. The adapter rank is selected to fit a hospital tablet GPU, while the base weights are quantized for local inference. A privacy gate prevents network calls and removes logging of raw patient text. The runtime batches only within a single device and uses a fixed answer budget to keep latency predictable. We evaluate MedQA accuracy, median on-device latency, peak memory, and a targeted drug-interaction error set. No external evidence store is consulted, so the system's clinical knowledge remains inside the adapted parameters.

3. Experiments

PocketClinic reaches 51.0% MedQA accuracy, approximately 80% of CuraLM-70B's 63.8%. Median latency is 210 ms, peak memory is 5.8 GB, and the full evaluation runs offline. In the targeted interaction set, PocketClinic fabricates an interaction in 12.7% of cases versus 7.1% for the larger system. The privacy gate incurs no measurable network overhead because the system makes no network calls.

System	MedQA accuracy	Latency	Memory
CuraLM-70B	63.8%	--	Data-center
PocketClinic	51.0%	210 ms	5.8 GB

4. Conclusion

PocketClinic shows that LoRA-QA can bring a 3B Small Language Model to a private hospital tablet. The trade-off is clear: offline operation and low memory come with lower Medical QA accuracy and more fabricated drug interactions. Structured relational memory is a promising future safeguard.

References

[1] Brandt, S. (2023). CuraLM-70B: Adapting a Large Language Model to Clinical Text. Journal of Clinical AI (JCAI).

[2] Tanaka, M. (2024). LoRA-QA: Parameter-Efficient Fine-Tuning of Sub-8B Models for Question Answering. Workshop on Efficient Language Models (WELM).

[3] Raman, P. (2022). Relational Fact Lattices for Grounding Compact Clinical Text Generators. Clinical Informatics Letters (CIL).